Ultimate obstination, constructively

a machine-checked proof and its computational content

Abstract

This note accompanies an Agda formalisation of Coquand's *Une preuve directe du Théorème d'Ultime Obstination* (`min1.pdf`), a direct constructive proof of Colson's ultimate-obstination theorem. We recall the statement, sketch the proof at a high level, and explain its computational pay-off: the Scott-continuous extension of a primitive-recursive term is a computable object, so one can *evaluate* it, in particular at the infinite diagonal point $x = S^\omega(\bot)$. We illustrate this with a primitive-recursive min, which returns $\bot$ on $(S^\omega(\bot), S^\omega(\bot))$ — obstination in one line. Every statement below is machine-checked with `agda -safe -without-K -exact-split`, with no postulates, holes, or pragmas.

1 The domain and primitive recursion

Let D be the domain of *lazy naturals*: its elements are the complete numbers $S^k(0)$, the incomplete finite ones $S^k(\bot)$ (with $S^0(\bot) = \bot$), and the single infinite element $S^\omega(\bot)$. The order is generated by $S^k(\bot) \leq S^\omega(\bot)$ and $S^k(\bot) \leq S^l(x)$ for $k \leq l$; the elements $S^k(0)$ and $S^\omega(\bot)$ are maximal. Write $F \subseteq D$ for the finite elements.

PR is the usual syntax of primitive-recursive functions: the constant 0, the projections, the successor S, and closure under composition and primitive recursion

$$f(0, Y) = g(Y), \qquad f(S\,x, Y) = h(x, f(x, Y), Y).$$

Each PR term of arity n denotes a monotone map $F^n \to F$, written $\mathrm{ev}(p)$; moreover every such map is *stable* (Berry): if A, B have a common upper bound then $f(A \wedge B) = f(A) \wedge f(B)$. Both facts are proved by a direct induction.

2 The property and the theorem

Definition 1 (Ultimate obstination)**.** *A monotone $f\colon F^n \to F$ satisfies the property at $A \in D^n$ when there is a finite $A_0 \leq A$ such that one of:*

1. *there is m with $f(X) = S^m(0)$ for all finite $X \geq A_0$;*
2. *there is m and a coordinate i with $A(i)$ incomplete finite, $A_0(i) = A(i)$, and $f(X) = S^m(\bot)$ for all finite X with $X(i) = A_0(i)$ and $X[i] \geq A_0[i]$;*
3. *there is a coordinate i with $A(i) = S^\omega(\bot)$, an integer k with $A_0(i) = S^k(\bot)$, and a numeric φ (constant or strictly increasing from k) with $f(X) = S^{\varphi(m)}(\bot)$ for all finite X with $X(i) = S^m(\bot)$, $k \leq m$, $A_0[i] \leq X[i]$.*

The existential and disjunctive content is read intuitionistically. We write $\mathrm{UO}\, f\, A$ *for a witness, and say* f satisfies ultimate obstination *if it does so at every point.*

In the formalisation this is a three-constructor datatype over the three cases (file `Property.agda`).

Proposition 1 (Colson; Coquand's note)**.** *Every element of* PR *satisfies ultimate obstination.*

In Agda (`Prop1.agda`):

$$\texttt{prop1} : (p : \mathrm{PR})\,(A : \mathrm{Tup}) \to \mathrm{Wf}\, p\, (\mathrm{length}\, A) \to \mathrm{UO}\,(\mathrm{ev}\, p)\, A,$$

where $\mathrm{Wf}\, p\, n$ records that p is well-formed of arity n (e.g. a projection π_i needs $i < n$; $\mathrm{prec}\, g\, h$ at arity $n = m{+}1$ needs g of arity m and h of arity $m{+}2$).

Proof sketch. Induction on p.

- *Constants, projections, successor* satisfy the property outright (each is eventually constant or reads a single coordinate).
- *Composition.* The witness functions φ are closed under composition, so $g \circ \langle h_1, \ldots, h_k\rangle$ is obstinate whenever the parts are.
- *Primitive recursion* is the heart. For a finite first argument one recurs on its height. For the infinite first argument $x = S^\omega(\bot)$ one forms the Kleene approximation

  $$u_0 = \bot, \qquad u_{k+1} = \hat{h}(S^\omega(\bot), u_k, Y),$$

  which is increasing and, once $u_k = u_{k+1}$, constant thereafter. A constructive dichotomy decides, at each stage n, whether the sequence has already stabilised (giving a finite limit, hence case 1 or 2) or still satisfies $S^n(\bot) \le u_n$. In the second regime one uses that h is obstinate at $(S^\omega(\bot), S^\omega(\bot), Y)$ and *stable* to pin the growth to a single coordinate and read off a witness ψ (with ψ constant or strictly increasing), yielding case 3. Stability — not the stronger Kahn–Plotkin sequentiality — is exactly what excludes two coordinates independently driving f to infinity.

The recursion argument is carried out once over an abstract base/step pair; the induction then feeds it the *arity-guarded* interpretations of the sub-terms (padded to be total and obstinate) and transports the result back to the concrete interpreter. $\square$

3 Computational content: evaluating $\hat{f}$

The point of phrasing obstination intuitionistically is that a witness *carries the value* of the Scott-continuous extension. Reading off case 1/2/3 gives

$$\mathsf{val}(\mathrm{UO}\, f\, A) = \begin{cases} S^m(0) & \text{case 1,} \\ S^m(\bot) & \text{case 2,} \\ S^{\varphi(k)}(\bot) & \text{case 3, } \varphi \text{ constant,} \\ S^\omega(\bot) & \text{case 3, } \varphi \text{ strictly increasing,} \end{cases}$$

and this is the value $\hat{f}(A)$ of the extension (file `Property.agda`, `uoValue`).

Corollary 1 (Computability). *For every* PR *term f of arity n the extension $\hat{f}\colon D^n \to D$ is computable: $\hat{f}(A) = \mathsf{val}(\mathtt{prop1}\, f\, A)$ is a total, effective function of A.*

Since `prop1` and `val` are total, postulate-free Agda functions, their composition *is* the algorithm (file `Computable.agda`):

$$\mathtt{fhat}\ :\ (f : \mathrm{PR})(A : \mathrm{Tup}) \to \mathrm{Wf}\, f\, (\mathrm{length}\, A) \to D.$$

In particular one may evaluate at the *infinite diagonal* $x = S^\omega(\bot)$ in every coordinate; call this $\mathtt{fhat\text{-}diag}\, f\, n$. These are not symbolic answers: for concrete f they reduce, inside the type-checker, to a concrete element of D.

4 pred **and** min **at** $S^\omega(\bot)$

Take the standard primitive-recursive terms (file `PredMin.agda`):

$$\begin{array}{ll}
\mathrm{pred} = \mathrm{prec}\ 0\ \pi_0 & \mathrm{pred}(0) = 0,\ \mathrm{pred}(S\,x) = x, \\
\mathrm{sub} = \mathrm{prec}\ \pi_0\ (\mathrm{pred} \circ \pi_1) & \mathrm{sub}(y, x) = x \dot{-} y, \\
\mathrm{min} = \mathrm{sub} \circ \langle \mathrm{sub} \circ \langle \pi_1, \pi_0 \rangle,\ \pi_0 \rangle & \mathrm{min}(x, y) = x \dot{-} (x \dot{-} y), \\
\mathrm{add} = \mathrm{prec}\ \pi_0\ (S \circ \pi_1) & \mathrm{add}(0, y) = y,\ \mathrm{add}(S\,x, y) = S(\mathrm{add}(x, y)).
\end{array}$$

On total inputs these are genuinely pred, truncated subtraction, min, and add (checked by reduction: $\mathrm{min}(2,3) = 2$, $\mathrm{min}(3,1) = 1$, $\mathrm{add}(2,3) = 5$, etc.). Evaluating their extensions at $x = S^\omega(\bot)$ gives:

$$\widehat{\mathrm{pred}}(x) = S^\omega(\bot), \qquad \widehat{\mathrm{sub}}(x, x) = \bot, \qquad \boxed{\widehat{\mathrm{min}}(x, x) = \bot}\ \text{(not } x!), \qquad \widehat{\mathrm{add}}(x, x) = S^\omega(\bot).$$

Each equality holds *definitionally* in Agda (`refl`).

The reason is exactly ultimate obstination. Subtraction recurs on its first argument; at $S^\omega(\bot)$ there is no outermost successor to peel, so the Kleene sequence never rises above $\bot$ and $\widehat{\mathrm{sub}}(S^\omega(\bot), x) = \bot$. Since $\mathrm{min} = x \dot{-} (x \dot{-} y)$ is built from subtraction, it inherits this obstination and collapses to $\bot$ on $(S^\omega(\bot), S^\omega(\bot))$. No primitive-recursive algorithm for min can avoid this: it must fully consume an argument, and on $S^\omega(\bot)$ that computation diverges. This is Colson's theorem, made concrete and machine-checked.

Addition is the instructive positive contrast. It, too, recurs on its first argument, but its step function is the *successor*, so the Kleene sequence $u_0 = \bot$, $u_{k+1} = S(u_k)$ is $u_k = S^k(\bot)$, strictly increasing with supremum $S^\omega(\bot)$. This is case 3 with φ strictly increasing, so $\widehat{\mathrm{add}}(S^\omega(\bot), S^\omega(\bot)) = S^\omega(\bot)$: infinity plus infinity is infinity. The denotation faithfully tracks the algorithm — min obstinately diverges to $\bot$, while add climbs to the infinite element.

Files. `Prop1.agda` (Proposition 1), `Property.agda` (the property and `val`), `Computable.agda` (the computability corollary), `PredMin.agda` (the pred/ min example), and the recursion chain `USeq*`, `PrecInf*`, `PrecBot*`, `PrecAll`. Source note: `min1.pdf`.